AICoaching

CSE 360, Fall 2026 (C) · Team 24: Arush Ugal, Karan Patel, Anirudh B, Pranjal Ganvir, Moksha Dev
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Purpose

TP1 built validation: a lesson field is refused when it is missing, too long, or badly formed. Coaching is a second layer that runs only after a field has passed validation. It accepts the input and then advises the contributor how to make the lesson more useful to the people who will search for it later: spelling, specificity, whether the fix is actually stated, and whether the keywords and category match what other lessons use.

This plan comes from each member's HW2 evaluation of LessonShare. LessonShare showed four AI functions worth adopting (an interview that asks one follow-up question at a time, takeaway extraction, a writing coach, and a one-line summary) and one obligation that comes with them: a consent step before any text leaves the application. The five fields below are the five a contributor types into the TP2 lesson editor; the sixth attribute the editor stores, `contributor`, is taken from the login and is never typed, so it needs no coaching.

Prioritized list of coached input fields

Priority reflects how much a weak value in the field reduces the value of a stored lesson to a later reader.

#	Field	What goes wrong without coaching	Owner	Primary tool
1	Description	The field readers rely on most. Misspellings defeat the search, and a vague description ("it broke") records that something happened but not what.	Arush Ugal	LanguageTool
2	Resolution	A missing or one-line resolution ("fixed it") leaves the reader with the problem and no fix.	Moksha Dev	Apache OpenNLP
3	Title	A generic title ("Bug fix") is useless in a list of search results.	Karan Patel	Groq API
4	Keywords	Contributors choose different words for the same idea ("db", "database", "H2"), so a keyword search misses lessons.	Anirudh B	KeyBERT
5	Category	Free-text categories fragment ("DB stuff", "Database", "Databases"), so the category filter splits lessons that belong together.	Pranjal Ganvir	Sentence Transformers

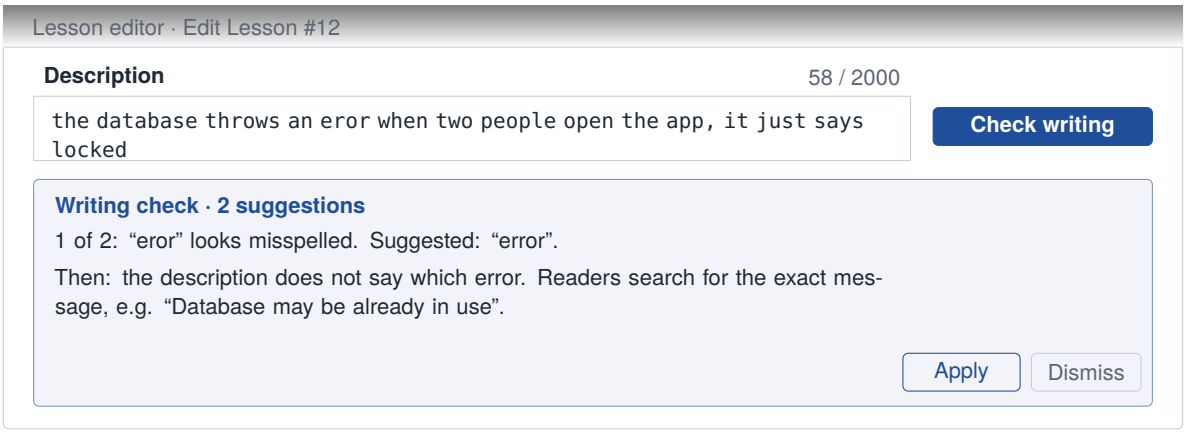
Workload. There are five fields and five members, so each member owns exactly one field: its mockup, concept of operations, requirements, tool evaluation, and, in TP3, its implementation and tests.

Requirements that apply to every field

C-1	Coaching runs only after the field has passed LessonInputPolicy validation, and never replaces it.
C-2	Coaching never blocks Save Lesson. Every suggestion can be dismissed.
C-3	The stored text changes only when the contributor presses Apply on a suggestion.
C-4	Before a field's text is sent to any service outside the application, the contributor is told which service receives it and must agree once; the choice is recorded with the account and can be withdrawn (LessonShare function 15).
C-5	If a coaching tool is unavailable or too slow (over 3 seconds), the editor works exactly as it does without coaching and shows no error.
C-6	At most one suggestion per field is shown at a time.

1 Description

Attribute `description` · Owner: **Arush Ugal**



Concept of operations

The contributor writes a description and presses Check writing. The text is checked by the LanguageTool library running inside the application, so nothing is sent anywhere. Spelling and grammar matches are shown one at a time, each with its replacement and an Apply button. After the language matches are handled, an optional second check, offered only to contributors who have given consent (C-4), sends the description to the same LLM writing coach LessonShare uses; it returns one sentence on what a reader would still need, for example the exact error message or the step that triggers the problem. Pressing Save Lesson at any point saves what is in the field.

Prioritized requirements

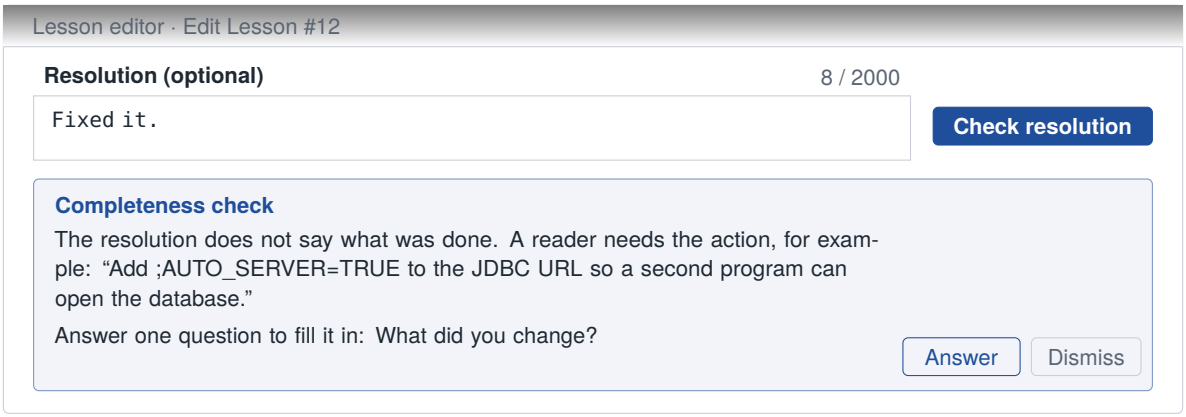
D-1 Must	Detect misspelled words and offer LanguageTool's first replacement.
D-2 Must	Never send the description outside the application for spelling and grammar; use the local library, not the public HTTP API.
D-3 Should	Detect grammar problems that change meaning (missing verb, wrong word) and offer the replacement LanguageTool gives.
D-4 Should	Ignore technical terms already used as keywords in stored lessons ("JDBC", "JavaFX"), so they are not reported as misspellings.

D-5 Could	With consent, ask the LLM writing coach for one sentence naming what is missing.

Tool	**LanguageTool** (Java library `languagetool-core`), open source under LGPL-2.1, runs offline inside a Java application.
Why	Java library, so it fits the Foundations code without a second runtime; local, so D-2 holds. The public HTTP API is not used: it is limited to 20 requests per IP per minute and its terms ask that automated use run a private instance.
Sources	Java API: `https://dev.languagetool.org/java-api` Source and licence: `https://github.com/languagetool-org/languagetool` HTTP API limits: `https://dev.languagetool.org/public-http-api`
LessonShare	Function 12, AI writing coach (feedback on a draft without blocking the save), used for D-5.

2 Resolution

Attribute `resolution` · Owner: **Moksha Dev**



Concept of operations

When the contributor leaves the Resolution field, or presses Check resolution, the text is split into sentences and tagged with parts of speech by Apache OpenNLP, locally. A resolution passes the completeness check when at least one sentence contains an action: a verb in base form at the start of a clause ("add", "set", "use", "restart") or a past-tense verb with an object ("changed the URL"). A resolution with no action, such as "Fixed it." or "Works now.", gets the panel above. An empty resolution is valid (the field is optional), so it gets no warning; instead the panel offers to ask one question, in the style of LessonShare's interview, when the lesson is next opened by its contributor.

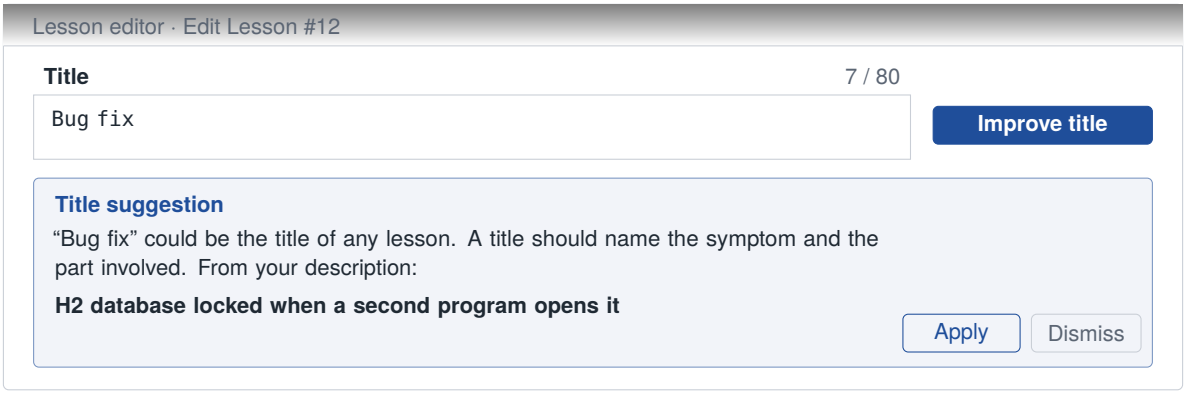
Prioritized requirements

R-1 Must	Report a non-empty resolution that contains no action, with an example of one.
R-2 Must	Do not report a resolution that contains an action, whatever its length.
R-3 Should	Do not warn about an empty resolution; offer the follow-up question instead.
R-4 Should	Once dismissed for a lesson, do not show the same panel again for that lesson in the same session.
R-5 Could	With consent, let the contributor answer the follow-up question and draft a resolution from the answer for them to edit.

Tool	**Apache OpenNLP** with its pre-trained English sentence detector and part-of-speech models, Apache-2.0, pure Java, runs offline.
Why	The check needs sentences and verb tags, not text generation; a local tagger answers it in milliseconds with no data leaving the application.
Sources	Manual: `https://opennlp.apache.org/docs/` Models: `https://opennlp.apache.org/models.html` Source and licence: `https://github.com/apache/opennlp`
LessonShare	Function 10, brainstorm interview (one focused follow-up question at a time), used for R-3 and R-5.

3 Title

Attribute `title` · Owner: **Karan Patel**



Concept of operations

The title is checked locally first: it is reported as generic when it has fewer than four words and every word is on a short list of generic terms (“bug”, “fix”, “error”, “problem”, “issue”, “help”, “question”), or when it matches the title of an existing lesson exactly. For a generic title, and only with consent, the description is sent to an LLM through the Groq API with an instruction to return one title of at most 80 characters that names the symptom and the component. This is LessonShare’s one-line summary used as a title generator. The suggestion is shown with Apply; it is checked against the 80-character limit before it is offered.

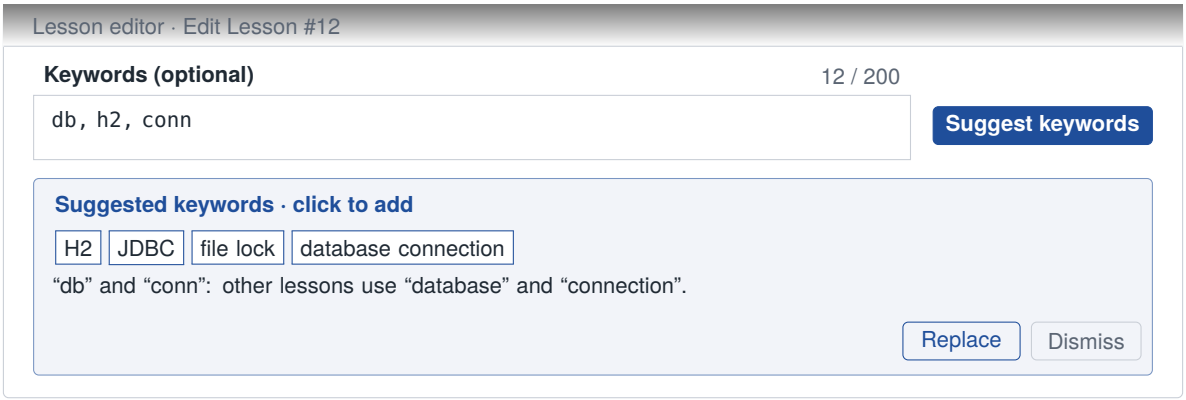
Prioritized requirements

T-1 Must	Report a title made only of generic terms, and a title identical to an existing lesson’s title.
T-2 Must	Any suggested title passes LessonInputPolicy.validateTitle (at most 80 characters) before it is shown.
T-3 Should	With consent, suggest a title generated from the description.
T-4 Should	Without consent, or if the service fails, still show the T-1 advice, with no suggestion.
T-5 Could	Offer up to three suggestions, one at a time, if the first is dismissed.

Tool	**Groq API**, free plan, with an open-weight Llama model; the service LessonShare already calls.
Why	Turning a description into a short title is a generation task, which a rule cannot do. The free plan's rate limits are far above what one contributor editing a title needs.
Sources	Quick start: `https://console.groq.com/docs/quickstart` Free plan limits: `https://console.groq.com/docs/rate-limits`
LessonShare	Function 13, one-line summary of a post; Function 15, consent gating (C-4).

4 Keywords

Attribute `keywords` · Owner: **Anirudh B**



Concept of operations

Pressing Suggest keywords runs KeyBERT on the title and description. KeyBERT embeds the text and its candidate words and phrases with the all-MiniLM-L6-v2 model and keeps the five phrases closest in meaning to the whole text. Each suggestion is shown as a chip; clicking it appends it to the field in the standard "first, second" form. Separately, every keyword already typed is compared with the keywords used by stored lessons; when a stored keyword is the long form of a typed abbreviation, a Replace action is offered. The model runs on the team's machine as a small local helper, so no text is sent to a third party and C-4 does not apply.

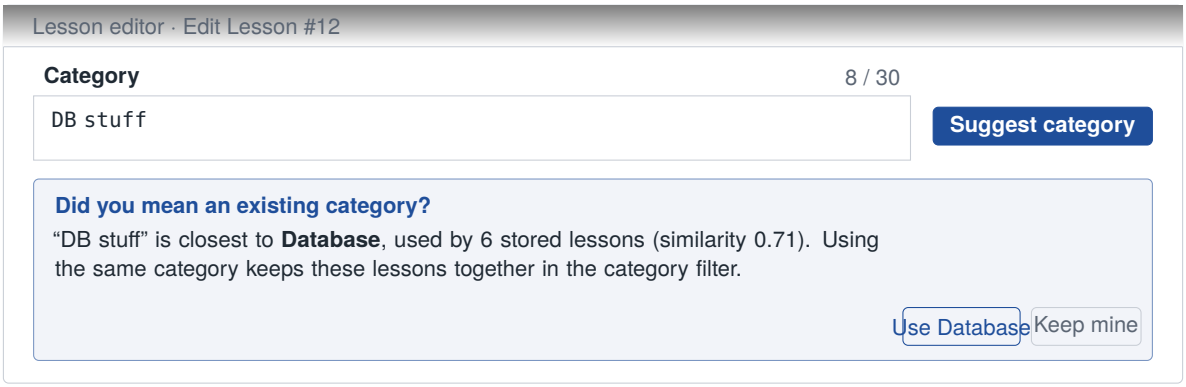
Prioritized requirements

K-1 Must	Suggest at most five keywords drawn from the title and description.
K-2 Must	Never suggest a keyword already in the field, ignoring case, and never let the field exceed the CS-11 limits (10 keywords, 30 characters each, 200 in total).
K-3 Should	Prefer keywords already used by stored lessons when two candidates mean the same thing, so searches find both lessons.
K-4 Should	Offer to replace an abbreviation with the form stored lessons use.
K-5 Could	Prefer two-word technical phrases ("file lock") to single common words.

Tool	**KeyBERT**, MIT licence, keyword extraction with BERT embeddings, run locally with the all-MiniLM-L6-v2 model.
Why	Unsupervised: it needs no training data, which matters while the lesson store is small. K-5 maps onto its keyphrase length setting.
Sources	Documentation: `https://maartengr.github.io/KeyBERT/` Source and licence: `https://github.com/MaartenGr/KeyBERT`
LessonShare	Function 14, AI keyword suggestions for a search; here applied when the lesson is written rather than when it is searched.

5 Category

Attribute `category` · Owner: **Pranjal Ganvir**



Concept of operations

When the contributor leaves the Category field, the typed category and every category returned by LessonsLearnedList.getCategories are turned into sentence embeddings with the all-MiniLM-L6-v2 model, run in the Java application through the Deep Java Library. The closest existing category by cosine similarity is offered when its score is at least 0.6 and it is not already the typed value. Below the threshold nothing is shown, because the contributor may be starting a genuinely new category. An exact match ignoring case ("database" for "Database") is corrected to the stored spelling with no model call.

Prioritized requirements

G-1 Must	Offer the stored spelling when the typed category differs from an existing one only in case or surrounding spaces.
G-2 Must	Offer the single closest existing category when its similarity is at least the threshold; show nothing otherwise.
G-3 Should	Show how many stored lessons use the suggested category.
G-4 Should	Keep the typed category when the contributor chooses Keep mine, and do not ask again for that lesson.
G-5 Could	Let staff adjust the threshold in TP3, after real categories have been collected.

Tool	**Sentence Transformers** model all-MiniLM-L6-v2 (Apache-2.0), run in Java with the **Deep Java Library** (Apache-2.0).
Why	The task is semantic similarity between short phrases, which is what this model is trained for; it is small enough to run on a laptop, so no category text leaves the application.
Sources	Model and licence: `https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2` Similarity method: `https://sbert.net/docs/sentence_transformer/usage/semantic_textual_similarity.html` Running it from Java: `https://djl.ai/`
LessonShare	LessonShare offers no category coaching; this extends its keyword-suggestion idea (function 14) to the category field.

Sources

LessonShare v3a and each member's HW2 evaluation of it (functions numbered as in the evaluation). All other sources are listed with their URL in the section that uses them; every URL was checked on October 25, 2026.